00-09 Hall Effect

Wirewrap Board Input Connections

SIGNAL	PURPOSE	Connector Pin
DAC0 Out	0V to +10V input	J1-1
Analog GND	Analog GND tied to Pin J1-3 on WW Logic Board	2
Digital GND	Digital GND tied to Pin J1-2 on WW Logic Board	3
Water Interlock Set	Set Water Latch	4
Water Interlock Reset	Reset Water Latch	5
Water Interlock Wiper	GND	6
U2 Pin 2	Shunt (-)	7
U2 Pin 3	Shunt (+)	8
PC0	Sample Count Bit LSB	9
PC1	Sample Count Bit MSB	10
PA2	Sample Current Direction	11
PA3	Magnet Current Direction	12
PA4	PC Generated Magnet Overload	13
PA5	PC ALARM	14
PA6	Spare	15
PA7	Spare	16
Vacuum Gauge +	Monitor Vacuum	17
Vacuum Gauge -	Monitor Vacuum	18
Unasigned		19
-15V		20
+15V		21
+5V		22

Wirewrap Board Output Connections

SIGNAL	PURPOSE	
Shunt Monitor	Monitor current, to PCI Card ACH0	J1-A
U1 Analog GND at R6	Analog GND to PCI Card ACH1	B
PB0	Vacuum monitor, Vacuum O.K. to PC	C
PB1	Water monitor, Water O.K. to PC	D
Input Voltage to H.P. Supplies		E
Input Ground Reference to H.P. Supplies		F
PB2 Magnet Shunted		H
Q9 Drain IRFZ44N n channel depletion mode to J11-A		N
Magnet Current Direction Forward - Green		J2-1
Ground		J2-2
Magnet Current Direction Reverse - Red		J2-3
Ground		J2-4
Magnet Current Direction Off - Yellow		J2-5
+5V		J2-6
Sample Current Direction Forward - Green		J2-7
Ground		J2-8
Sample Current Direction Reverse - Red		J2-9
Ground		J2-10

Wirewrap Board Output Connections

SIGNAL	PURPOSE	
Vacuum O.K.	Green	J3-3
Vacuum not O.K.	Red	J3-2
Ground		J3-1
Water O.K.	Green	J4-3
Water not O.K.	Red	J4-2
Ground		J4-1
Sample Pin A	Red	J5-3
Sample Pin A	Green	J5-2
Ground		J5-1
Sample Pin B	Red	J6-3
Sample Pin B	Green	J6-2
Ground		J6-1
Sample Pin C	Red	J7-3
Sample Pin C	Green	J7-2
Ground		J7-1
Sample Pin D	Red	J8-3
Sample Pin D	Green	J8-2
Ground		J8-1

Board To Board Connections

U11c Pin 14	Drive for Q1	J9-1
U11d Pin 13	Drive for Q2	J9-2
U11e Pin 12	Drive for Q3	J9-3
U11f Pin 11	Drive for Q4	J9-4
U11b Pin 15	Drive for Q5	J9-5
U10b Pin 15	Drive for Q6	J9-6
U10a Pin 16	Drive for Q7	J9-7
U11a Pin 16	Drive for Alarm	J9-8
Unasigned		J9-9
Unasigned		J9-10
Unasigned		J9-11
Unasigned		J9-12
Unasigned		J9-13
U10 Pin 9	+24V From Relay Board	J9-14

Relay Board Input Connections

J9-1	FROM LOGIC BOARD	Drive for Q1	J10-1
J9-2		Drive for Q2	J10-2
J9-3		Drive for Q3	J10-3
J9-4		Drive for Q4	J10-4
J9-5		Drive for Q5	J10-5
J9-6		Drive for Q6	J10-6
J9-7		Drive for Q7	J10-7
J9-8		Drive for Alarm	J10-8
J9-9			J10-9
J9-10			J10-10
J9-11			J10-11
J9-12			J10-12
J9-13			J10-13
J9-14		+24V	J10-14

Base Drive for Q7 to J1-N	J11-A
Unasigned	J11-B
Unasigned	J11-C
Unasigned	J11-D
Unasigned	J11-E
Unasigned	J11-F
Unasigned	J11-H
Unasigned	J11-J
Unasigned	J11-K
Unasigned	J11-L
Unasigned	J11-M
V3	J11-N
V3 Return	J11-P
V4	J11-R
V4 Return	J11-S
I1	J11-T
I2	J11-U
+5V	J11-V
+5V SUPPLY Return	J11W
Unasigned	J11-X
+24V SUPPLY Return	J11-Y
+24V	J11-Z

Relay Board Output Connections

A		J12-1
B		J12-2
C		J12-3
D		J12-4
Blinking Red LED Cathode	Magnet Shunted	J13-1
Blinking Red LED Anode	Magnet Shunted	J13-2
Relay 19 Return		J14-1
Relay 19 Drive		J14-2
Relay 18 Return		J14-3
Relay 18 Drive		J14-4

Terminal Block Connections

1	H.P. Supply Output (+)	J15-1
2	H.P. Supply Output (-)	J15-2
3	D10 Cathode Jumper to Pin 1	J15-3
4	D10 Anode Jumper to Pin2	J15-4
5	unasigned	J15-5
6	unasigned	J15-6
7	Input to H.P. Supply J1-E	J15-7
8	Input to H.P. Supply J1-F	J15-8
9	Magnet (+)	J15-9
10	Magnet (-)	J15-10

Water Interlock Connections

A				
B				
C				
D				
E				
F				
G				
H				
I				
J				
K				
L	Set Water Latch	- to J1-4	Red	J16-L
M	Reset Water Latch	- to J1-5	Wht	J16-M
N	GND	- to J1-6	Blk	J16-N

Cryostat Connections

A	Sample A From J12-1	J17-A
B	Sample B From J12-2	J17-B
C	Sample C From J12-3	J17-C
D	Sample D From J12-4	J17-D
E	Heater (-)	J17-E
F	Heater (+)	J17-F
G	+V	J17-G
H	+I	J17-H
J	-I	J17-J
K	-V	J17-K

Line 1 = 25 Pin Male D – Sub connector back of box

Pin 1	From J1-A to ACH0	Pin 1 from PCI 1200 Card
Pin 2	From J1-B to ACH01	Pin 2 from PCI 1200 Card
Pin 3	Not Connected	Pin 3 from PCI 1200 Card
Pin 4	Not Connected	Pin 4 from PCI 1200 Card
Pin 5	Not Connected	Pin 5 from PCI 1200 Card
Pin 6	Not Connected	Pin 6 from PCI 1200 Card
Pin 7	Not Connected	Pin 7 from PCI 1200 Card
Pin 8	Not Connected	Pin 8 from PCI 1200 Card
Pin 9	Not Connected	Pin 9 from PCI 1200 Card
Pin 10	From J1-1 to DAC0 Out	Pin 10 from PCI 1200 Card
Pin 11	From J1-2 to ANA GND	Pin 11 tied to DIG GND on WW Logic Brd
Pin 12	Not Connected	Pin 12 from PCI 1200 Card
Pin 13	From J1-3 to DIG GND	Pin 13 tied to ANA GND on WW Logic Brd
Pin 14	Not Connected	Pin 14 from PCI 1200 Card
Pin 15	Not Connected	Pin 15 from PCI 1200 Card
Pin 16	From J1-11 to PA2	Pin 16 from PCI 1200 Card
Pin 17	From J1-12 to PA3	Pin 17 from PCI 1200 Card
Pin 18	From J1-13 to PA4	Pin 18 from PCI 1200 Card
Pin 19	From J1-14 to PA5	Pin 19 from PCI 1200 Card
Pin 20	From J1-15 to PA6	Pin 20 from PCI 1200 Card
Pin 21	From J1-16 to PA7	Pin 21 from PCI 1200 Card
Pin 22	From J1-C to PB0	Pin 22 from PCI 1200 Card
Pin 23	From J1-D to PB1	Pin 23 from PCI 1200 Card
Pin 24	From J1-H to PB2	Pin 24 from PCI 1200 Card
Pin 25	AC Ground	

Line 2 = 25 Pin Female D – Sub connector back of box

Pin 1	+5V Supply from SHE Box	Pin 25 from PCI 1200 Card
Pin 2	Not Connected	Pin 26 from PCI 1200 Card
Pin 3	Not Connected	Pin 27 from PCI 1200 Card
Pin 4	Not Connected	Pin 28 from PCI 1200 Card
Pin 5	Not Connected	Pin 29 from PCI 1200 Card
Pin 6	To J1-9 From PC0	Pin 30 from PCI 1200 Card
Pin 7	To J1-10 From PC1	Pin 31 from PCI 1200 Card
Pin 8	Not Connected	Pin 32 from PCI 1200 Card
Pin 9	Not Connected	Pin 33 from PCI 1200 Card
Pin 10	Not Connected	Pin 34 from PCI 1200 Card
Pin 11	Not Connected	Pin 35 from PCI 1200 Card
Pin 12	Not Connected	Pin 36 from PCI 1200 Card
Pin 13	Not Connected	Pin 37 from PCI 1200 Card
Pin 14	Not Connected	Pin 38 from PCI 1200 Card
Pin 15	Not Connected	Pin 39 from PCI 1200 Card
Pin 16	Not Connected	Pin 40 from PCI 1200 Card
Pin 17	Not Connected	Pin 41 from PCI 1200 Card
Pin 18	Not Connected	Pin 42 from PCI 1200 Card
Pin 19	Not Connected	Pin 43 from PCI 1200 Card
Pin 20	Not Connected	Pin 44 from PCI 1200 Card
Pin 21	Not Connected	Pin 45 from PCI 1200 Card
Pin 22	Not Connected	Pin 46 from PCI 1200 Card
Pin 23	Not Connected	Pin 47 from PCI 1200 Card
Pin 24	Not Connected	Pin 48 from PCI 1200 Card
Pin 25	AC Ground	Pin 49 from PCI 1200 Card